

Data Structures



Submitted to

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Lab Tasks

Lab Task 01

Code:

```
int main()
{
    int x, start, stop;
    int sum=0;
    cout<<"Enter X:";
    cin>>x;

    cout<<"Enter starting value:";
    cin>>start;

    cout<<"Enter stopping value:";
    cin>>stop;

    for(int i=start;i<=stop;i++)
    {
        sum= sum+x*x;
    }

    cout<<"Sum="<<sum;
    return 0;
}
```

Result

```
Enter X:4
Enter starting value:2
Enter stopping value:5
Sum=64
-----
Process exited after 11.68 seconds with return value 0
Press any key to continue . . . |
```

Lab Task 02

Code:

```
#include <iostream>

using namespace std;

class ArrayList {
private:
    int arr[100];
    int size;

public:
    ArrayList() {
        size = 0;
    }

    // 1. Insert at end
    void insertEnd(int val) {
        if (size < 100) {
            arr[size] = val;
```

```
        size++;  
        cout << "Inserted " << val << " at end\n";  
    } else {  
        cout << "List full!\n";  
    }  
}
```

// 2. Insert at start

```
void insertStart(int val) {  
    if (size < 100) {  
        for (int i = size; i > 0; i--)  
            arr[i] = arr[i - 1];  
        arr[0] = val;  
        size++;  
        cout << "Inserted " << val << " at start\n";  
    } else {  
        cout << "List full!\n";  
    }  
}
```

// 3. Insert after specific value

```
void insertAfter(int after, int val) {  
    int pos = -1;  
    for (int i = 0; i < size; i++) {  
        if (arr[i] == after) {  
            pos = i;  
            break;  
        }  
    }  
}
```

```

if (pos == -1) {
    cout << after << " not found!\n";
    return;
}
if (size < 100) {
    for (int i = size; i > pos + 1; i--)
        arr[i] = arr[i - 1];
    arr[pos + 1] = val;
    size++;
    cout << "Inserted " << val << " after " << after << endl;
} else {
    cout << "List full!\n";
}
}

```

// 4. Insert before specific value

```

void insertBefore(int before, int val) {
    int pos = -1;
    for (int i = 0; i < size; i++) {
        if (arr[i] == before) {
            pos = i;
            break;
        }
    }
    if (pos == -1) {
        cout << before << " not found!\n";
        return;
    }
    if (size < 100) {

```

```
        for (int i = size; i > pos; i--)
            arr[i] = arr[i - 1];
        arr[pos] = val;
        size++;
        cout << "Inserted " << val << " before " << before << endl;
    } else {
        cout << "List full!\n";
    }
}
```

// 5. Display

```
void display() {
    if (size == 0) {
        cout << "List is empty!\n";
        return;
    }
    cout << "List: ";
    for (int i = 0; i < size; i++)
        cout << arr[i] << " ";
    cout << endl;
}
```

// 6. Delete from end

```
void deleteEnd() {
    if (size == 0) {
        cout << "List empty!\n";
        return;
    }
    cout << "Deleted " << arr[size - 1] << " from end\n";
}
```

```
    size--;  
}
```

// 7. Delete from start

```
void deleteStart() {  
    if (size == 0) {  
        cout << "List empty!\n";  
        return;  
    }  
    cout << "Deleted " << arr[0] << " from start\n";  
    for (int i = 0; i < size - 1; i++)  
        arr[i] = arr[i + 1];  
    size--;  
}
```

// 8. Delete specific value

```
void deleteVal(int val) {  
    int pos = -1;  
    for (int i = 0; i < size; i++) {  
        if (arr[i] == val) {  
            pos = i;  
            break;  
        }  
    }  
    if (pos == -1) {  
        cout << val << " not found!\n";  
        return;  
    }  
    for (int i = pos; i < size - 1; i++)
```

```
        arr[i] = arr[i + 1];  
    size--;  
    cout << "Deleted " << val << endl;  
}
```

// Lab Task 3: Linear Search using while loop

```
void linearSearch(int val) {  
    int i = 0;  
    int found = 0;  
  
    while (i < size) {  
        if (arr[i] == val) {  
            cout << val << " found at index " << i << endl;  
            found = 1;  
            break;  
        }  
        i++;  
    }  
  
    if (found == 0)  
        cout << val << " not found!\n";  
}  
};
```

```
int main() {  
    ArrayList list;  
    int choice, val, target;  
  
    do {
```

```
cout << "\n===== MENU =====" << endl;
cout << "1. Insert at end" << endl;
cout << "2. Insert at start" << endl;
cout << "3. Insert after specific value" << endl;
cout << "4. Insert before specific value" << endl;
cout << "5. Display list" << endl;
cout << "6. Delete from end" << endl;
cout << "7. Delete from start" << endl;
cout << "8. Delete specific value" << endl;
cout << "9. Linear Search (while loop)" << endl;
cout << "10. Exit" << endl;
cout << "Choice: ";
cin >> choice;
```

```
switch (choice) {
```

```
    case 1:
```

```
        cout << "Enter value: ";
```

```
        cin >> val;
```

```
        list.insertEnd(val);
```

```
        break;
```

```
    case 2:
```

```
        cout << "Enter value: ";
```

```
        cin >> val;
```

```
        list.insertStart(val);
```

```
        break;
```

```
    case 3:
```

```
        cout << "Enter after value: ";
```

```
cin >> target;  
cout << "Enter new value: ";  
cin >> val;  
list.insertAfter(target, val);  
break;
```

case 4:

```
cout << "Enter before value: ";  
cin >> target;  
cout << "Enter new value: ";  
cin >> val;  
list.insertBefore(target, val);  
break;
```

case 5:

```
list.display();  
break;
```

case 6:

```
list.deleteEnd();  
break;
```

case 7:

```
list.deleteStart();  
break;
```

case 8:

```
cout << "Enter value to delete: ";  
cin >> val;
```

```
list.deleteVal(val);  
break;  
  
case 9:  
    cout << "Enter value to search: ";  
    cin >> val;  
    list.linearSearch(val);  
    break;  
  
case 10:  
    cout << "Exiting...\n";  
    break;  
  
default:  
    cout << "Invalid choice!\n";  
    }  
} while (choice != 10);  
  
return 0;  
}
```

Result

```
C:\Users\Laraib\Documents\C  ×  +  v

===== MENU =====
1. Insert at end
2. Insert at start
3. Insert after specific value
4. Insert before specific value
5. Display list
6. Delete from end
7. Delete from start
8. Delete specific value
9. Linear Search (while loop)
10. Exit
Choice: 1
Enter value: 2
Inserted 2 at end

===== MENU =====
1. Insert at end
2. Insert at start
3. Insert after specific value
4. Insert before specific value
5. Display list
6. Delete from end
7. Delete from start
8. Delete specific value
9. Linear Search (while loop)
10. Exit
Choice: |
```

Lab Task 02

Code:

```
#include <iostream>

using namespace std;

int main() {
    int arr[100], size, search, i;

    cout << "Enter number of elements: ";
    cin >> size;

    cout << "Enter " << size << " elements: ";
    for (i = 0; i < size; i++)
        cin >> arr[i];

    cout << "Enter value to search: ";
    cin >> search;

    i = 0;
    while (i < size && arr[i] != search)
        i++;

    if (i < size)
        cout << search << " found at index " << i << endl;
    else
        cout << search << " not found!" << endl;

    return 0;
}
```

Code:

```
Enter number of elements: 4
Enter 4 elements: 2 3 5 7
Enter value to search: 5
5 found at index 2

-----
Process exited after 20.25 seconds with return value 0
Press any key to continue . . . |
```